MATH 231 – Linear Algebra I

The Matrix Unit, Part 7 ▪ The Determinant in General

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About this document. This is Part 7 of the matrix unit, Sections 35–39. Section numbers run continuously through the unit (Part 2's opening table lists them all).

Assumed. The 2×2 determinant: its definition (§14), its properties P1–P11 (§22), and its meaning as an area (§23). Rank and null space (§12, §31–§32). Orthogonal matrices (§20) and the rotations of $\mathbb{R}^3$ (§25).

What you should be able to do afterwards. Compute minors and cofactors; compute the determinant of a 3×3 or 4×4 matrix by cofactor expansion, choosing a good row or column; find principal and leading principal minors; use the properties of the determinant to compute it without expanding; and prove that the determinant of a triangular matrix is the product of its diagonal and that an orthogonal matrix has determinant ±1.

For a 2×2 matrix the number $ad-bc$ did three jobs. It measured how the matrix scales area, its sign recorded whether the plane was flipped, and it was nonzero exactly when the matrix had full rank. We want a number with the same three powers for an $n\times n$ matrix: a volume factor in $\mathbb{R}^3$, and the same test of rank in every dimension. It is built recursively, out of determinants one size smaller, with $ad-bc$ as the first step.

35. Minors and cofactors

35.1. Minors

Definition 35.1 (Minor)**.** Let $\mathbf{A}\in\mathbb{R}^{n\times n}$ with $n\geq2$. Deleting row i and column j of $\mathbf{A}$ leaves an $(n-1)\times(n-1)$ matrix, written $\mathbf{A}_{ij}$. Its determinant

$$M_{ij}=\det(\mathbf{A}_{ij})$$

is the (i,j) **minor** of $\mathbf{A}$.

The definition is recursive: a minor of an $n\times n$ matrix is the determinant of a matrix one size smaller. For a 3×3 matrix the minors are 2×2 determinants, which we know how to compute.

Example 35.1. For $\mathbf{A}=\begin{bmatrix}1&2&3\\4&5&6\\7&8&10\end{bmatrix}$, the three minors along the first row are

$$M_{11}=\det\begin{bmatrix}5&6\\8&10\end{bmatrix}=2,\qquad M_{12}=\det\begin{bmatrix}4&6\\7&10\end{bmatrix}=-2,\qquad M_{13}=\det\begin{bmatrix}4&5\\7&8\end{bmatrix}=-3.$$

For M_{12}, row 1 and column 2 were crossed out, leaving the four entries $4,6,7,10$.

35.2. Cofactors

Definition 35.2 (Cofactor)**.** The (i,j) **cofactor** of $\mathbf{A}$ is the minor with a sign attached:

$$C_{ij}=(-1)^{i+j}M_{ij}.$$

The sign $(-1)^{i+j}$ is $+$ when $i+j$ is even and $-$ when it is odd, which makes a checkerboard starting with $+$ in the top-left corner:

$$\begin{bmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

In Example 35.1, $C_{11} = +2$, $C_{12} = -(-2) = 2$, $C_{13} = +(-3) = -3$.

35.3. The 2×2 case, seen again

Agree that the determinant of a 1×1 matrix $[\,a\,]$ is the number a itself. Then for $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, deleting row 1 and column 1 leaves $[\,d\,]$, and deleting row 1 and column 2 leaves $[\,c\,]$, so $C_{11} = d$ and $C_{12} = -c$, and

$$aC_{11} + bC_{12} = ad - bc = \det \mathbf{A}.$$

The 2×2 formula already has the shape "each entry of the first row, times its cofactor, added up". That shape is the general definition.

36. The Laplace expansion

36.1. The definition

Definition 36.1 (Determinant)**.** For $\mathbf{A} \in \mathbb{R}^{n \times n}$: if $n = 1$, $\det[\,a\,] = a$. If $n \geq 2$,

$$\boxed{\det \mathbf{A} = \sum_{j=1}^{n} a_{1j} C_{1j} = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}.}$$

This is the **Laplace expansion**, or **cofactor expansion**, along the first row.

Each cofactor is a determinant of size $n-1$, computed by the same rule, until the sizes reach 2×2 or 1×1.

Example 36.1. For the matrix of Example 35.1,

$$\det \mathbf{A} = 1 \cdot C_{11} + 2 \cdot C_{12} + 3 \cdot C_{13} = 1(2) + 2(2) + 3(-3) = -3.$$

36.2. The 3×3 formula

Written out for a general 3×3 matrix, the definition gives

$$\det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}),$$

a sum of six products. Each product takes exactly one entry from each row and one from each column. That is true in every size, and it is the starting point of the closed formula (with permutations) that more advanced courses use.

36.3. Any row, any column

The first row has no special status.

Theorem 36.1 (Expansion along any row or column)**.** *For $\mathbf{A} \in \mathbb{R}^{n\times n}$ and any fixed i and j,*

$$\det \mathbf{A} = \sum_{k=1}^{n} a_{ik}C_{ik} \quad (\textit{along row } i), \qquad \det \mathbf{A} = \sum_{k=1}^{n} a_{kj}C_{kj} \quad (\textit{down column } j).$$

All $2n$ expansions give the same number. The proof needs the closed formula just mentioned and is beyond this course; we take the theorem as given. It can at least be checked.

Example 36.2. Expand the same $\mathbf{A}$ down its first column. The cofactors needed are $C_{11} = 2$, $C_{21} = -\det \begin{bmatrix} 2 & 3 \\ 8 & 10 \end{bmatrix} = -(20-24) = 4$ and $C_{31} = +\det \begin{bmatrix} 2 & 3 \\ 5 & 6 \end{bmatrix} = 12 - 15 = -3$, so

$$\det \mathbf{A} = 1(2) + 4(4) + 7(-3) = 2 + 16 - 21 = -3. \quad \checkmark$$

36.4. Choosing wisely

Every zero entry in the chosen row or column kills a whole term, cofactor and all. So expand along the row or column with the most zeros.

Example 36.3. $\mathbf{B} = \begin{bmatrix} 2 & 0 & 1 \\ 3 & 0 & 4 \\ 1 & 5 & 2 \end{bmatrix}$. Column 2 has two zeros, so only $b_{32} = 5$ contributes:

$$\det \mathbf{B} = 5\, C_{32} = 5 \cdot (-1)^{3+2} \det \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} = -5(8-3) = -25.$$

Along the first row instead: $2 \det \begin{bmatrix} 0 & 4 \\ 5 & 2 \end{bmatrix} - 0 + 1 \det \begin{bmatrix} 3 & 0 \\ 1 & 5 \end{bmatrix} = 2(-20) + 15 = -25$. Same answer, more work.

Example 36.4. A 4×4 example, where the choice matters more:

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}.$$

Row 2 has a single nonzero entry, $c_{22} = 3$, with sign $(-1)^{2+2} = +$:

$$\det \mathbf{C} = 3 \det \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix}.$$

The 3×3 determinant has a single nonzero entry in its middle row, 2, again with sign $+$: it equals $2 \det \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} = 2(1-2) = -2$. So $\det \mathbf{C} = 3(-2) = -6$.

36.5. The cost

Expanding an $n \times n$ determinant creates n determinants of size $n-1$, each of which creates $n-1$ of size $n-2$, and so on: about $n \cdot (n-1) \cdots 2 = n!$ products in all. For $n = 10$ that is about 3.6 million; for $n = 20$, about 2.4×10^{18}, more than a fast computer can do in a year. Cofactor expansion is a definition and a tool for small or sparse matrices, not an algorithm for large ones. Part 8 computes determinants by elimination, with about $\frac{2}{3}n^3$ operations: 5,000-odd for $n = 20$.

36.6. Volume

For a 3×3 matrix, $|\det \mathbf{A}|$ is the volume of the parallelepiped (the slanted box) with edges the three columns, and every volume is multiplied by $|\det \mathbf{A}|$ when $\mathbf{A}$ is applied. This is §23 one dimension up; we state it without proof. The sign records orientation: positive if the columns, in order, are arranged like $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ (a right-handed arrangement), negative if like their mirror image. For a box with edges along the axes, $\det \operatorname{diag}(a, b, c) = abc$, the familiar length times width times height (§39.2).

37. Principal minors

37.1. The definition

A minor deletes one row and one column. A *principal* minor keeps the rows and columns in matching sets.

Definition 37.1 (Principal and leading principal minors)**.** Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ and choose a set S of indices from $\{1, \ldots, n\}$. Keeping the rows *and* the columns numbered by S gives a **principal submatrix**; its determinant is a **principal minor**. When $S = \{1, 2, \ldots, k\}$, the submatrix is the top-left $k \times k$ corner, and its determinant Δ_k is the **k-th leading principal minor**.

So $\Delta_1 = a_{11}$, $\Delta_2 = \det \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$, and so on up to $\Delta_n = \det \mathbf{A}$. A principal submatrix always has its own diagonal on the diagonal of $\mathbf{A}$, which is the point of keeping the same indices.

37.2. An example

Example 37.1. $\mathbf{A} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 4 \end{bmatrix}$.

- Size 1 ($S = \{1\}, \{2\}, \{3\}$): the diagonal entries $2, 3, 4$.
- Size 2: $S = \{1, 2\}$ gives $\det \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} = 5$; $S = \{1, 3\}$ gives $\det \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} = 8$; $S = \{2, 3\}$ gives $\det \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix} = 11$.
- Size 3: $\det \mathbf{A} = 2(12 - 1) - 1(4 - 0) + 0 = 18$.

The leading principal minors are $\Delta_1 = 2$, $\Delta_2 = 5$, $\Delta_3 = 18$.

Note that the size-1 principal minors add up to the trace (§9). The size-2 ones and the size-n one (the determinant) also have a meaning, which the next unit uncovers.

37.3. What they are for

A symmetric matrix $\mathbf{A}$ is called **positive definite** if $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} > 0$ for every $\mathbf{x} \neq \mathbf{0}$. (The expression $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x}$ appeared in §20.) Such matrices are the matrix version of a positive number. They are the Hessians of functions shaped like a bowl, and they appear in least squares in Part 9. The leading principal minors detect them.

Theorem 37.1 (Sylvester's criterion, 2×2)**.** *A symmetric* $\mathbf{A} = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ *is positive definite if and only if* $\Delta_1 = a > 0$ *and* $\Delta_2 = ac - b^2 > 0$.

Proof. Here $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} = ax_1^2 + 2bx_1x_2 + cx_2^2$. If $a > 0$, complete the square:

$$ax_1^2 + 2bx_1x_2 + cx_2^2 = a\Big(x_1 + \frac{b}{a}x_2\Big)^2 + \frac{ac - b^2}{a}\,x_2^2.$$

If also $ac - b^2 > 0$, both terms are ≥ 0, and both are 0 only if $x_2 = 0$ and then $x_1 = 0$. Conversely, if $\mathbf{A}$ is positive definite then $\mathbf{x} = \mathbf{e}_1$ gives $a > 0$, and $\mathbf{x} = [\,-b, a\,]$ makes the first term vanish and gives $a(ac - b^2) > 0$, so $ac - b^2 > 0$. □

For $n \times n$ the same holds: a symmetric matrix is positive definite exactly when all n leading principal minors are positive. We state this without proof. Example 37.1's matrix $(2, 5, 18)$ is positive definite. Applied to a Hessian (the vector calculus unit, §15), this is the multivariable second-derivative test for a minimum.

38. Properties of the determinant

These are the laws that make the determinant computable without expanding, and they generalize §22 property by property. They are labelled D1–D10, the way Part 1 labelled the laws of multiplication M1–M6. Throughout, $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$.

38.1. The laws

	Law	Status
D1	$\det \mathbf{I}_n = 1$	proved
D2	$\det \mathbf{A}^{\mathsf{T}} = \det \mathbf{A}$	proved, from Theorem 36.1
D3	a zero row gives $\det \mathbf{A} = 0$	proved
D4	scaling one row by α scales det by α; so $\det(\alpha\mathbf{A}) = \alpha^n \det \mathbf{A}$	proved
D5	swapping two rows changes the sign	stated
D6	two equal rows give $\det \mathbf{A} = 0$	proved, from D5
D7	det is linear in each row separately	proved
D8	adding a multiple of one row to another leaves det unchanged	proved, from D6–D7
D9	$\det(\mathbf{A}\mathbf{B}) = \det \mathbf{A} \det \mathbf{B}$	stated ($n = 2$ in §22)
D10	$\det \mathbf{A} \neq 0 \iff \operatorname{rank} \mathbf{A} = n \iff \operatorname{null}[\mathbf{A}] = \{\mathbf{0}\}$	half here, half in §42.6

By D2, every law about rows is also a law about columns.

38.2. The proofs

D1. Expand $\mathbf{I}_n$ along its first row: the only nonzero term is $1 \cdot \det \mathbf{I}_{n-1}$. Repeat down to $\det \mathbf{I}_1 = 1$.

D2. By induction on n; for $n = 1$ there is nothing to prove. Expand $\mathbf{A}^\mathsf{T}$ along its first row. Its entries are $a_{11}, a_{21}, \ldots, a_{n1}$, the first column of $\mathbf{A}$. Deleting row 1 and column j of $\mathbf{A}^\mathsf{T}$ leaves the transpose of $\mathbf{A}_{j1}$, whose determinant equals $\det \mathbf{A}_{j1}$ by the induction hypothesis. So the expansion of $\mathbf{A}^\mathsf{T}$ along row 1 is term by term the expansion of $\mathbf{A}$ down column 1, which is $\det \mathbf{A}$ by Theorem 36.1.

D3. Expand along the zero row: every term has a factor 0.

D4. Expand along the scaled row. Every term $a_{ik}C_{ik}$ becomes $(\alpha a_{ik})C_{ik}$, while the cofactors C_{ik} are unchanged, because they never use row i. Scaling all n rows multiplies by α n times. So $\det(2\mathbf{I}_3) = 8$: doubling every length in space multiplies every volume by 8.

D5 is stated. On Example 35.1's matrix, swapping rows 1 and 2 gives $\det \begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 7 & 8 & 10 \end{bmatrix} = 4(-4) - 5(-11) + 6(-6) = 3 = -(-3)$.

D6. Swapping the two equal rows leaves $\mathbf{A}$ unchanged but, by D5, changes the sign of the determinant. So $\det \mathbf{A} = -\det \mathbf{A}$, and $\det \mathbf{A} = 0$.

D7. If row i is $\alpha\mathbf{u} + \beta\mathbf{v}$, expand along row i: each term is $(\alpha u_k + \beta v_k)C_{ik}$, and the cofactors do not involve row i. Splitting the sum gives α times the determinant with row i equal to $\mathbf{u}$ plus β times the one with row i equal to $\mathbf{v}$.

D8. Replace row i by $\mathbf{r}_i + k\mathbf{r}_j$ with $j \neq i$. By D7 the new determinant is $\det \mathbf{A}$ plus k times the determinant of the matrix whose rows i and j are both $\mathbf{r}_j$, which is 0 by D6.

$$\det \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 10 \end{bmatrix} = 1(-30 + 48) - 2(0 + 42) + 3(0 + 21) = -3,$$

the determinant of Example 35.1's matrix after subtracting 4 times row 1 from row 2. Unchanged.

D10, first half. *If the columns are dependent,* $\det \mathbf{A} = 0$. Suppose column k is a combination of the others, $\mathbf{a}_k = \sum_{j \neq k} c_j \mathbf{a}_j$. By D7 for columns, $\det \mathbf{A}$ is $\sum_{j \neq k} c_j$ times the determinant of the matrix with $\mathbf{a}_j$ written in column k. Each of those has two equal columns, so is 0 by D6 for columns. Hence $\operatorname{rank} \mathbf{A} < n$ forces $\det \mathbf{A} = 0$. The converse, that $\operatorname{rank} \mathbf{A} = n$ forces $\det \mathbf{A} \neq 0$, comes out of elimination in §42.6.

38.3. What the row operations do

Three of the laws describe what happens to the determinant under the three simplest ways of changing rows:

Operation on the rows	**Effect on** det
swap two rows	multiply by -1 (D5)
multiply one row by $\alpha \neq 0$	multiply by α (D4)
add a multiple of one row to another	no change (D8)

These are exactly the three operations of Gaussian elimination in Part 8. That is why elimination computes determinants cheaply.

A common confusion (No law for sums). There is a law for products, D9, and none for sums: in general $\det(\mathbf{A}+\mathbf{B}) \neq \det\mathbf{A} + \det\mathbf{B}$. Already $\det(\mathbf{I}_3+\mathbf{I}_3) = 8$ while $\det\mathbf{I}_3 + \det\mathbf{I}_3 = 2$.

39. Determinants of special matrices

39.1. Triangular matrices

Theorem 39.1. *If $\mathbf{A} \in \mathbb{R}^{n\times n}$ is triangular (upper or lower), its determinant is the product of its diagonal entries:*

$$\det\mathbf{A} = a_{11}a_{22}\cdots a_{nn}.$$

Proof. Lower triangular first, by induction on n; the case $n = 1$ is the definition. Row 1 of a lower triangular matrix is $[\,a_{11}, 0, \ldots, 0\,]$, so expanding along it leaves one term:

$$\det\mathbf{A} = a_{11}C_{11} = a_{11}\det\mathbf{A}_{11}.$$

Deleting row 1 and column 1 of a lower triangular matrix leaves a lower triangular matrix of size $n-1$ with diagonal $a_{22}, \ldots, a_{nn}$, whose determinant is $a_{22}\cdots a_{nn}$ by the induction hypothesis. For an upper triangular $\mathbf{A}$, the transpose is lower triangular with the same diagonal, and D2 finishes the proof. □

Example 39.1. $\det\begin{bmatrix}2 & 5 & -1\\ 0 & 3 & 7\\ 0 & 0 & 4\end{bmatrix} = 2\cdot 3\cdot 4 = 24$. The entries above the diagonal play no role at all.

In particular, a triangular matrix has determinant 0 exactly when some diagonal entry is 0. In Part 8 that becomes the condition under which a triangular system can be solved by back substitution.

39.2. Diagonal matrices

A diagonal matrix is triangular, so $\det\operatorname{diag}(d_1,\ldots,d_n) = d_1d_2\cdots d_n$. Geometrically, $\operatorname{diag}(d_1,\ldots,d_n)$ stretches the k-th axis by d_k, and a unit cube becomes a box with edges $|d_1|,\ldots,|d_n|$: the volume factor is the product. With every $d_k = 1$ this is D1 again.

39.3. Orthogonal matrices

Theorem 39.2. *If $\mathbf{Q} \in \mathbb{R}^{n\times n}$ is orthogonal, then $\det\mathbf{Q} = \pm 1$.*

Proof. From $\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{I}$, by D9, D2 and D1, $(\det\mathbf{Q})^2 = \det\mathbf{Q}^\mathsf{T}\det\mathbf{Q} = \det(\mathbf{Q}^\mathsf{T}\mathbf{Q}) = \det\mathbf{I} = 1$. □

An orthogonal matrix preserves lengths and angles (§20), so it preserves volume, and the theorem confirms the factor is 1; the sign records whether space is turned over. The rotations of §25 have determinant $+1$:

$$\det\mathbf{R}_z(\theta) = \det\begin{bmatrix}\cos\theta & -\sin\theta & 0\\ \sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix} = 1\cdot\det\mathbf{R}_\theta = 1,$$

expanding along the third row. The same works for $\mathbf{R}_x$ (along the first row) and $\mathbf{R}_y$ (along the second). The swap of two coordinates, $\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$, is $\mathbf{I}_3$ with two rows exchanged, so by D5 and D1 its determinant is -1: a reflection across the plane $x = y$.

A common confusion (The converse is false). $\det \mathbf{A} = \pm 1$ does not make $\mathbf{A}$ orthogonal. The shear $\begin{bmatrix}1&1\\0&1\end{bmatrix}$ has determinant 1 and changes lengths (§21.4).

39.4. Projections

Theorem 39.3. *If* $\mathbf{P}^2 = \mathbf{P}$ *and* $\mathbf{P} \neq \mathbf{I}$*, then* $\det \mathbf{P} = 0$.

Proof. Since $\mathbf{P} \neq \mathbf{I}$, there is a vector $\mathbf{x}$ with $\mathbf{z} = (\mathbf{I} - \mathbf{P})\mathbf{x} \neq \mathbf{0}$. Then $\mathbf{P}\mathbf{z} = (\mathbf{P} - \mathbf{P}^2)\mathbf{x} = \mathbf{0}$, so null[$\mathbf{P}$] contains $\mathbf{z} \neq \mathbf{0}$. By Theorem 31.2 of Part 6 the columns of $\mathbf{P}$ are dependent, and by the first half of D10, $\det \mathbf{P} = 0$. □

So a projection that is not the identity always loses volume entirely: it flattens space onto a lower-dimensional subspace.

39.5. Practice

P1. Compute $\det \begin{bmatrix}2&-1&0\\1&3&2\\0&1&4\end{bmatrix}$ along the first row, and again down the first column.

P2. Compute the determinant of the upper triangular matrix with diagonal $2, -1, 3, \frac{1}{2}$ (and anything above the diagonal).

P3. $\mathbf{A}$ is 3×3 with $\det \mathbf{A} = 5$. Find $\det(2\mathbf{A})$, $\det(-\mathbf{A})$, $\det(\mathbf{A}^\mathsf{T}\mathbf{A})$, and the determinant of the matrix obtained by swapping two rows of $\mathbf{A}$.

P4. Find the leading principal minors of $\begin{bmatrix}4&2&0\\2&3&1\\0&1&2\end{bmatrix}$. Is it positive definite?

P5. Explain without computing why $\det \begin{bmatrix}1&2&3\\2&4&6\\1&0&1\end{bmatrix} = 0$.

P6. For which k is $\det \begin{bmatrix}1&1&1\\1&2&k\\1&4&k^2\end{bmatrix} = 0$? Explain the answer by looking at the columns.

Answers

P1. Along row 1: $2(12-2) - (-1)(4-0) + 0 = 20 + 4 = 24$. Down column 1: $2(12-2) - 1 \cdot \det \begin{bmatrix}-1&0\\1&4\end{bmatrix} + 0 = 20 - (-4) = 24$. ✓

P2. $2 \cdot (-1) \cdot 3 \cdot \frac{1}{2} = -3$.

P3. $2^3 \cdot 5 = 40$; $(-1)^3 \cdot 5 = -5$; $5 \cdot 5 = 25$; -5.

P4. $\Delta_1 = 4$, $\Delta_2 = 12 - 4 = 8$, $\Delta_3 = 4(6-1) - 2(4-0) + 0 = 12$. All positive, so it is positive definite.

P5. Row 2 is twice row 1. By D8, subtracting twice row 1 from row 2 does not change the determinant and produces a zero row, so by D3 the determinant is 0.

P6. Along the first row: $(2k^2 - 4k) - (k^2 - k) + (4 - 2) = k^2 - 3k + 2 = (k-1)(k-2)$, so $k = 1$ or $k = 2$. The columns are $[\,1, x, x^2\,]$ for $x = 1, 2, k$; when k equals 1 or 2, two columns coincide.

Looking ahead. Part 8 finally answers the question Part 1 left open: how to solve $\mathbf{Ax} = \mathbf{b}$. The method uses the three row operations of §38.3 to make the matrix triangular, and as a by-product it computes the determinant with a few thousand operations instead of $n!$.